UNIT 8

# DUAL NATURE OF RADIATION AND MATTER

*"If quantum mechanics has not profoundly shocked you, you have not understood it yet"*
– Neils Bohr

**In this unit, the students are exposed to**

- the phenomenon of electron emission and its types
- the observations of Hertz, Hallwachs and Lenard
- photoelectric effect and its laws
- the concept of quantization of energy
- photo cell and its applications
- particle nature of radiation
- the wave nature of matter
- de Broglie equation and de Broglie wavelength of electron
- the construction and working of electron microscope
- Davisson and Germer experiment
- X-rays and its production
- X-rays spectra and its types

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## 8.1 INTRODUCTION

We are familiar with the concepts of particle and wave in our everyday experience. Marble balls, grains of sand, atoms, electrons and so on are some examples of particles while the examples of waves are sea waves, ripples in a pond, sound waves and light waves.

Particle is a material object which is considered as a tiny concentration of matter (localized in space and time) whereas wave is a broad distribution of energy (not localized in space and time). They, both particles and waves, have the ability to carry energy and momentum from one place to another.

Classical physics which describes the motion of the macroscopic objects treats particles and waves as separate components of physical reality. The mechanics of particles and the optics of waves are traditionally independent subjects, each with its own experiments and principles.

Electromagnetic radiations are regarded as waves because they exhibit wave nature in phenomena such as interference, diffraction and polarization under some suitable circumstances. Similarly, under other circumstances like black body radiation and photo electric effect, electromagnetic radiations behave as though they consist of stream of particles.

When electrons, protons and other particles are discovered, they are considered as particles because they possess mass and



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